

Retail Sales Data Agent Task

Table Overview

Catalog Explorer > retail > retail_sales >

retail_sales_dataset

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OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Description

Created by the file upload UI

Filter columns...

AI generate

Column	Type	Popularity	Comment	Tags	Column masking	
Transaction ID	bigint					
Date	date					
Customer ID	string					
Gender	string					
Age	bigint					
Product Category	string					
Quantity	bigint					
Price per Unit	bigint					
Total Amount	bigint					

Table preview

Table +

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	
1	1	2023-11-24	CUST001	Male	34	Beauty	3	
2	2	2023-02-27	CUST002	Female	26	Clothing	2	
3	3	2023-01-13	CUST003	Male	50	Electronics	1	
4	4	2023-05-21	CUST004	Male	37	Clothing	1	
5	5	2023-05-06	CUST005	Male	30	Beauty	2	
6	6	2023-04-25	CUST006	Female	45	Beauty	1	
7	7	2023-03-13	CUST007	Male	46	Clothing	2	
8	8	2023-02-22	CUST008	Male	30	Electronics	4	
9	9	2023-12-13	CUST009	Male	63	Electronics	2	
10	10	2023-10-07	CUST010	Female	52	Clothing	4	

Table +

	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	IST001	Male	34	Beauty	3	50	150
2	IST002	Female	26	Clothing	2	500	1000
3	IST003	Male	50	Electronics	1	30	30
4	IST004	Male	37	Clothing	1	500	500
5	IST005	Male	30	Beauty	2	50	100
6	IST006	Female	45	Beauty	1	30	30
7	IST007	Male	46	Clothing	2	25	50
8	IST008	Male	30	Electronics	4	25	100
9	IST009	Male	63	Electronics	2	300	600
10	IST010	Female	52	Clothing	4	50	200

Connecting Data Agent to retail_sales_dataset

Connect your data

Genie Agents empower you to uncover meaningful insights from your data. Just upload your datasets, provide instructions, and simply ask your data questions.

For you All Metric views

All catalogs > retail > retail_sales 

  retail_sales_dataset

Selected:  retail_sales_dataset 

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Agent instructions

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General Instructions

Add general instructions on how you want this Genie Agent to behave.

🔧 Improve with Genie

Retail Sales Agent

Role: You are a retail sales agent whos' role is to answer questions based on the retail sales dataset as your source of data inorder to assist the business CEO understand and evaluate product perfomance, sales and customers to help grow the business. Also provide some business recommendations based on the results from analysis.

To explore: Compare the different categories, customers gender groups and time periods, answer all questions about sales, quantity sold, transactions, customers as well as the product categories and do all the necessary comparisons. Your answers must be clear and easy to understand and include numbers where applicable.

Limitations: Do not come up or invent anything that is not on the dataset, answer questions based on the dataset only.

Save

10 Questions and Answers

1. What are the total amount generated from all transactions?

Agent answer:

 Thought process >

Total Revenue: \$456,000

The total amount generated from all transactions in the retail sales dataset is **\$456,000**.¹

This represents the cumulative revenue across all product categories (Electronics, Beauty, and Clothing), customer segments, and transaction periods captured in the dataset.

Sources (1)

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2. Which product category sold the most quantity and how many they?

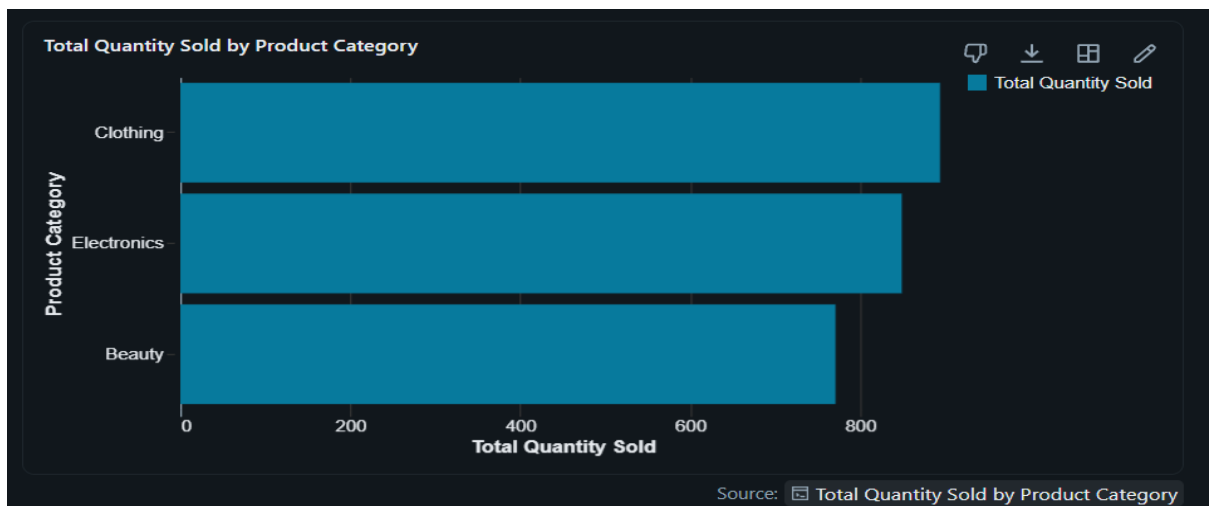
Agent answer:

Clothing Sold the Most: 894 Units

Clothing is the product category that sold the most quantity, with a total of **894 units** sold.¹

This is followed by Electronics with 849 units and Beauty with 771 units. Clothing leads by 45 units over Electronics, representing approximately 35.6% of all units sold across the three categories.

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Business Recommendation

Since Clothing is your top-selling category by volume, consider:

- **Expanding the clothing inventory** with more variety or trending styles to capitalize on customer demand
- **Analyzing which specific clothing items** drive the most sales to optimize stock levels
- **Cross-selling strategies** to boost Electronics and Beauty sales, which have lower volumes

Sources (1)

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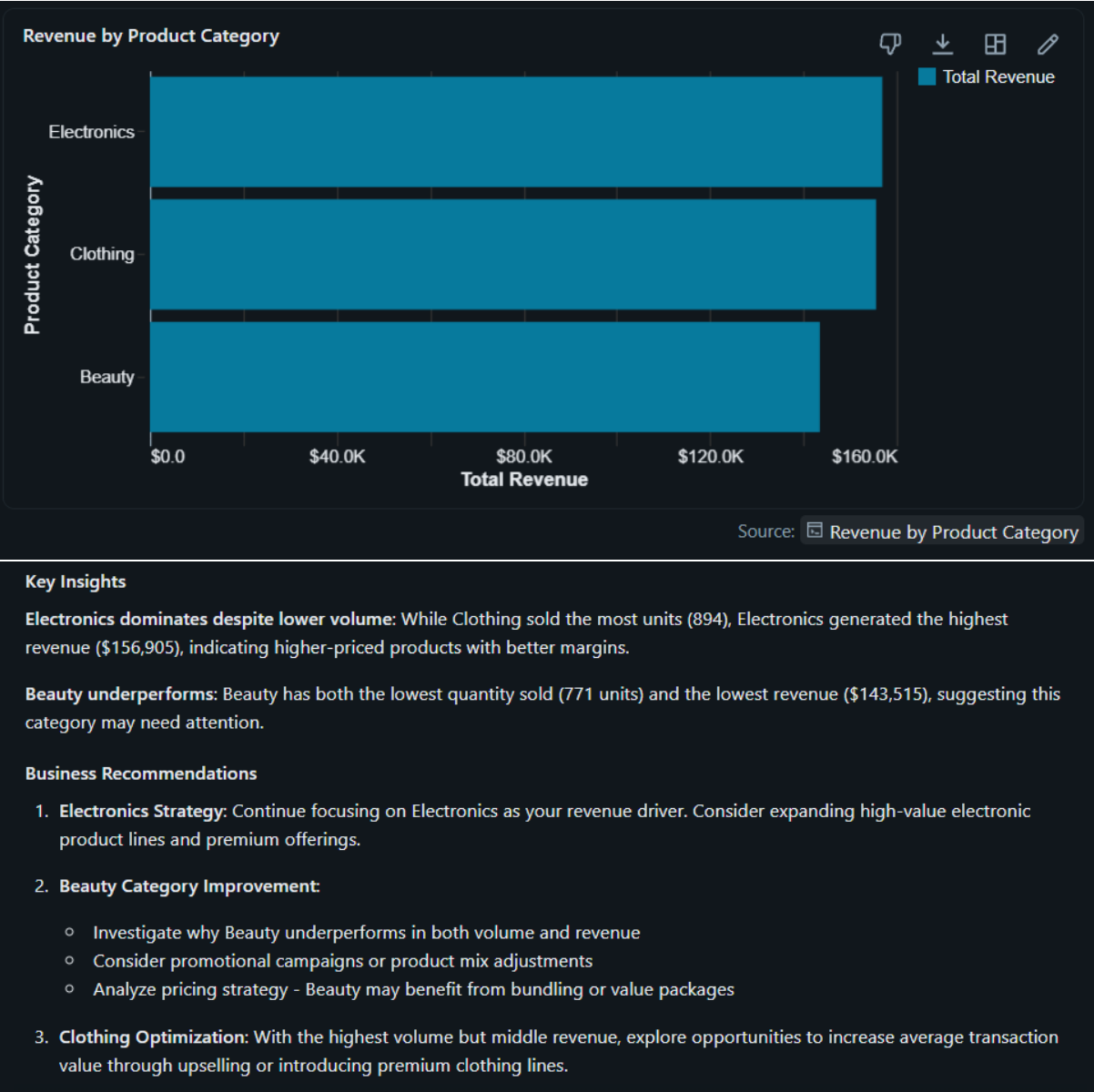
3. Which category made the most revenue and which one has the lowest revenue and how much was it?

Agent answer:

Electronics Leads Revenue at \$156,905; Beauty Has Lowest at \$143,515

Electronics generated the highest revenue with \$156,905, while Beauty had the lowest revenue at \$143,515. ¹

The revenue gap between the highest and lowest performing categories is \$13,390, representing approximately 9.3% difference. Clothing falls in the middle with \$155,580 in revenue, just \$1,325 behind Electronics.



4. Compare sales by gender, how much did each gender spend in total?

Agent answer:

Female Customers Spent More: \$232,840 vs \$223,160

Female customers spent a total of \$232,840, while male customers spent \$223,160. ¹

Female customers outspent male customers by \$9,680, representing approximately 4.3% higher total spending.

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Total Spending by Gender



Detailed Breakdown

Transaction Volume: Female customers made 510 transactions compared to 490 transactions by male customers—a difference of 20 transactions (4.1% more). ¹

Average Spending per Transaction: The spending patterns are remarkably similar:

- Female: \$456.55 per transaction
- Male: \$455.43 per transaction ²

This indicates that the higher total spending by female customers is driven primarily by **transaction volume** (more purchases) rather than higher spending per transaction.

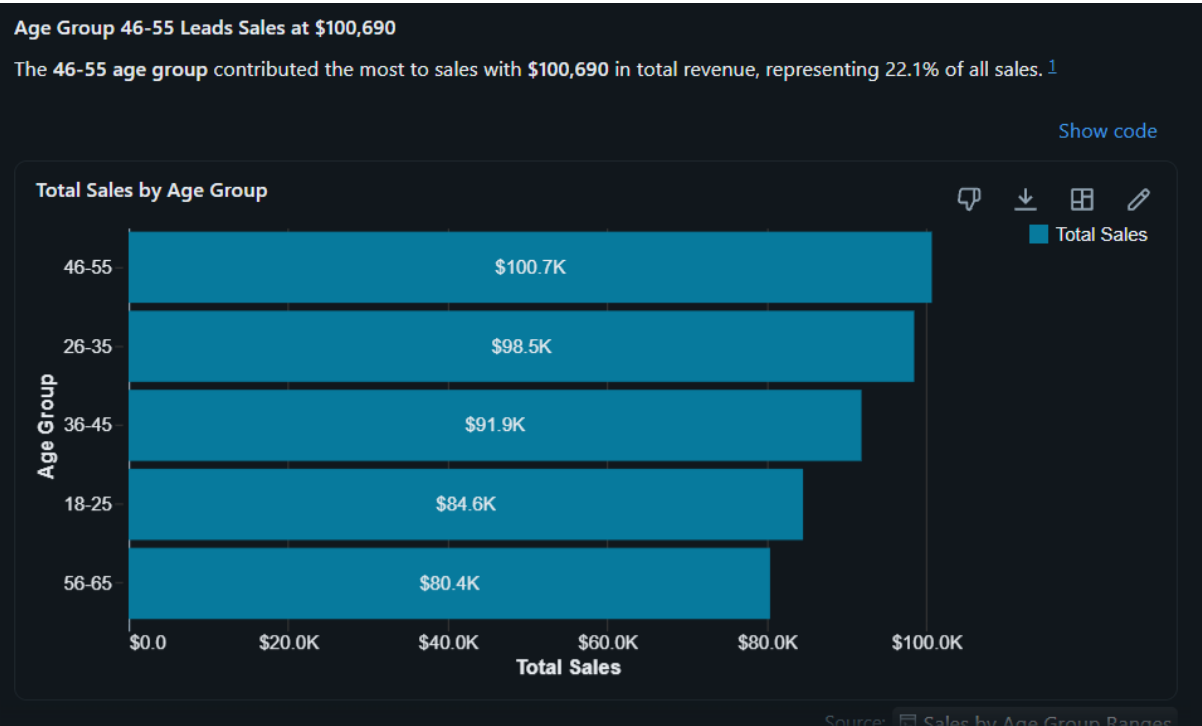
Business Recommendations

- Maintain Female Customer Engagement:** Since female customers drive slightly higher revenue through more frequent purchases, continue marketing efforts that resonate with this segment.
- Increase Male Customer Frequency:** Male customers spend nearly the same amount per transaction but make fewer purchases. Consider:
 - Targeted promotions to increase purchase frequency
 - Loyalty programs to encourage repeat visits
 - Personalized marketing campaigns

- Balanced Strategy:** With nearly equal spending per transaction across both genders, your product pricing and value proposition appeals equally to both segments—maintain this balance while working to increase overall transaction frequency.

5. Which age group contributed the most when it comes to sales?

Agent answer:



Sales Distribution by Age Group

Here's how each age group performed:

1. **46-55 years:** \$100,690 (229 transactions, 22.1% of total sales)
2. **26-35 years:** \$98,480 (205 transactions, 21.6% of total sales)
3. **36-45 years:** \$91,870 (202 transactions, 20.1% of total sales)
4. **18-25 years:** \$84,550 (169 transactions, 18.5% of total sales)
5. **56-65 years:** \$80,410 (195 transactions, 17.6% of total sales) ¹

Key Insights

Highest spending per transaction: The **18-25 age group** has the highest average spending per transaction at **\$500.30**, despite having the second-lowest total sales. This suggests younger customers make fewer but higher-value purchases. ¹

Most frequent shoppers: The **46-55 age group** leads with 229 transactions, showing both high frequency and solid per-transaction spending (\$439.69 average).

Lowest per-transaction spending: The **56-65 age group** has the lowest average at \$412.36 per transaction, indicating more frequent but smaller purchases.

Business Recommendations

1. **Prioritize 46-55 Age Group:** This is your core customer segment. Focus retention and loyalty programs on this demographic to maintain their high transaction frequency.

2. **Increase Frequency for 18-25:** Young adults spend the most per transaction but shop less frequently. Consider:
 - Targeted promotions to increase visit frequency
 - Subscription or membership programs
 - Social media engagement campaigns
3. **Grow 26-35 Segment:** This group is nearly tied with the top segment (\$98,480 vs \$100,690). Small improvements here could make them your #1 revenue driver.
4. **Engage Seniors (56-65):** While they have the lowest per-transaction spending, they shop frequently (195 transactions). Bundle deals or volume discounts could increase their basket size.

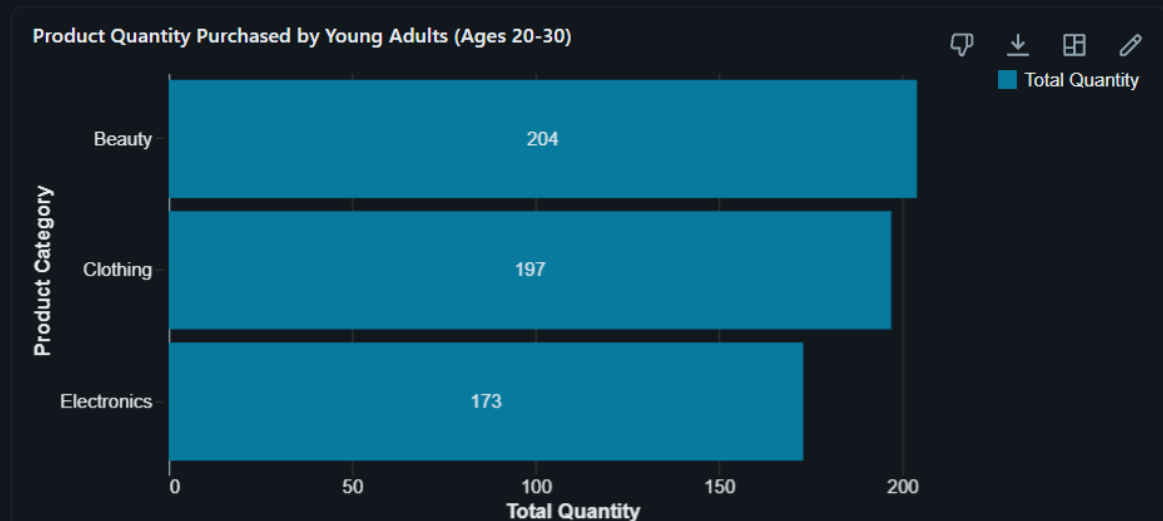
6. Which product do young people between ages 20-30 buy the most?

Agent answer:

Beauty Products Lead Among Young Adults: 204 Units

Young people between ages 20-30 buy **Beauty products** the most, purchasing a total of **204 units** across 81 transactions. ¹

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Product Preferences by Young Adults (Ages 20-30)

1. **Beauty:** 204 units (81 transactions, \$36,385 revenue)
2. **Clothing:** 197 units (78 transactions, \$41,855 revenue)
3. **Electronics:** 173 units (72 transactions, \$28,620 revenue) ¹

Key Insights

Clothing generates highest revenue despite lower volume: While Beauty leads in quantity, **Clothing generates the most revenue** from this age group at **\$41,855**, with an average of \$536.60 per transaction—the highest among all categories. ¹

Beauty is most popular by volume: Young adults make the most Beauty purchases (204 units, 81 transactions), indicating frequent, smaller purchases with an average of \$449.20 per transaction.

Electronics lags behind: Electronics has the lowest performance in this age group with only 173 units sold and \$28,620 in revenue, averaging \$397.50 per transaction.

Business Recommendations

1. **Expand Beauty Product Range:** Since Beauty is the most frequently purchased category by young adults, consider:
 - Introducing trendy or seasonal beauty products
 - Creating beauty bundles or subscription boxes
 - Leveraging social media marketing targeting this demographic
2. **Upsell Clothing:** Young adults already spend the most per transaction on Clothing (\$536.60). Opportunities include:
 - Premium clothing lines or limited editions
 - Style consultation services
 - Loyalty rewards for high-value clothing purchases
3. **Boost Electronics Appeal:** Electronics underperforms with this age group. Consider:
 - Tech accessories and gadgets popular with young adults (headphones, smart devices, gaming)
 - Student discounts or financing options
 - Bundle deals combining Electronics with other categories
4. **Cross-Category Promotions:** Create bundles that combine Beauty and Clothing products to increase basket size and leverage both categories' strengths.

7. Which month has highest revenue and lowest?, suggest what business can do to improve sales on lowest months.

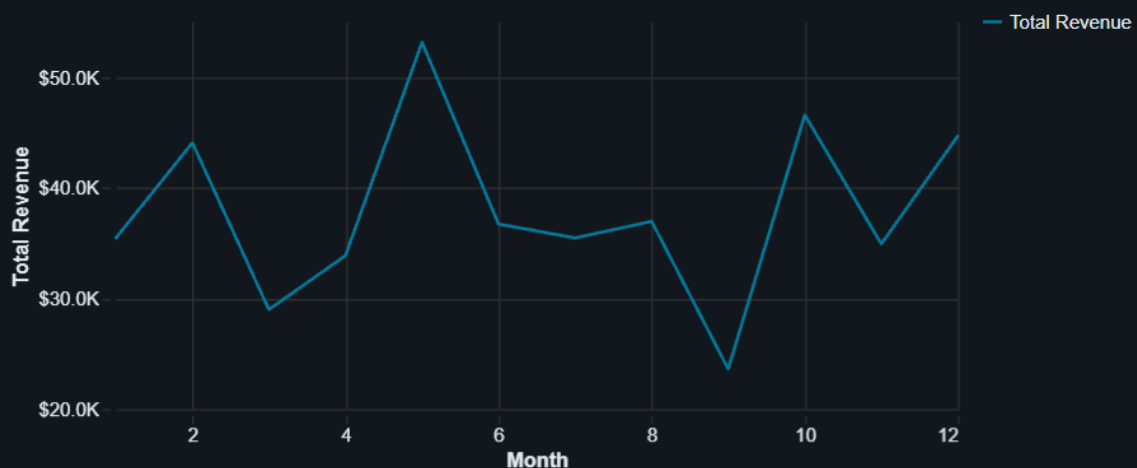
Agent answer:

May Had Highest Revenue (\$53,150); September Had Lowest (\$23,620)

May 2023 generated the highest revenue at \$53,150 with 105 transactions, while September 2023 had the lowest revenue at \$23,620 with only 65 transactions. ¹

The revenue gap between the best and worst months is \$29,530—a significant 125% difference, indicating substantial seasonal variation in sales performance.

Monthly Revenue Trend (2023)



Source: 2023 Monthly Revenue Analysis

Business Recommendations to Improve Low-Performing Months

1. Seasonal Promotions & Marketing Campaigns

For September (Back-to-School Season):

- Launch "Back to School" promotions targeting students and parents
- Bundle deals: Electronics + Clothing packages for students
- Student discount programs (10-15% off with valid student ID)
- Focus on categories popular with young adults (ages 20-30): Beauty and Clothing

For March (Spring Season):

- "Spring Refresh" campaign for Beauty and Clothing
- Spring fashion collections and beauty product launches
- "Spring Cleaning Sale" to clear winter inventory
- Easter/Spring holiday promotions

2. Loyalty & Retention Programs

- **Double Points Days:** Offer 2x loyalty points during slow months (March, September) to incentivize purchases
- **Exclusive Member Sales:** Early access to sales for loyalty members during these months
- **Birthday Month Rewards:** Since you have customer data, send personalized birthday offers to customers whose birthdays fall in low-performing months

3. Product-Specific Strategies

Electronics (severely underperformed in March with only \$3,380):

- Tech refresh promotions: "Upgrade Your Tech" campaigns
- Trade-in programs for old electronics
- New product launches timed for slow months
- Gaming and entertainment bundles

Clothing & Beauty:

- Seasonal transitions: Spring/Fall fashion previews
- "Buy 2 Get 1 Free" promotions
- Beauty sampling events or gift-with-purchase offers

4. Customer Engagement Tactics

- **Email Marketing Blitz:** Increase email frequency 2-3 weeks before slow months with personalized product recommendations
- **Flash Sales:** Weekly flash sales during September and March to create urgency
- **Social Media Contests:** Engagement campaigns with prizes to drive traffic
- **Referral Bonuses:** "Refer a Friend" programs with extra incentives during slow months

5. Inventory & Pricing Optimization

- **Strategic Discounting:** Plan 15-20% discounts on select items during slow months
- **Bundle Pricing:** Create attractive multi-category bundles at discounted rates
- **Clearance Events:** Use March and September to clear seasonal inventory

6. Target High-Value Segments

Focus on your best-performing demographics during slow months:

- **Age 46-55** (your top revenue group): Targeted promotions via email/direct mail
- **Female customers** (higher spenders): Exclusive women's shopping events
- **Age 26-35** (second-highest revenue): Digital-first campaigns on social media

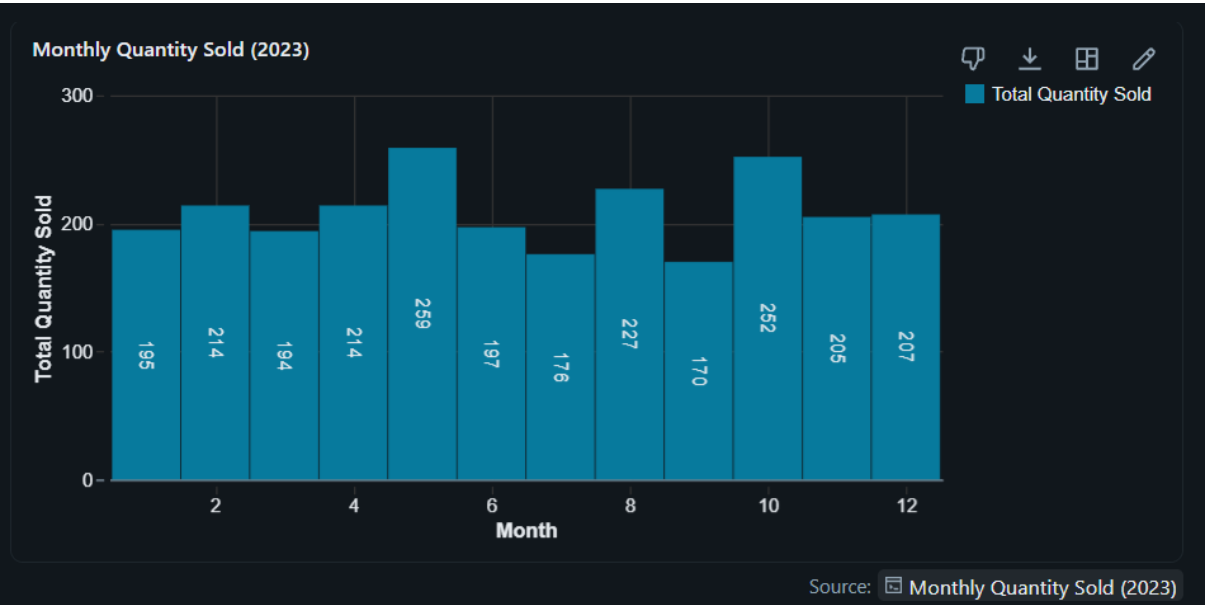
8. Which month has highest and lowest quantity sold and which category makes the most revenue on highest month?

Agent answer:

May Had Highest Quantity (259 Units); September Had Lowest (170 Units)

May 2023 sold the most quantity with 259 units, while September 2023 had the lowest with 170 units sold. ¹

The difference between the highest and lowest months is 89 units, representing a 52% gap in volume performance.



Monthly Quantity Performance

Top 3 Months by Quantity:

1. May: 259 units (105 transactions, \$53,150 revenue)
2. October: 252 units (96 transactions, \$46,580 revenue)
3. August: 227 units (94 transactions, \$36,960 revenue) ¹

Bottom 3 Months by Quantity:

1. September: 170 units (65 transactions, \$23,620 revenue)
2. July: 176 units (72 transactions, \$35,465 revenue)
3. March: 194 units (73 transactions, \$28,990 revenue) ¹

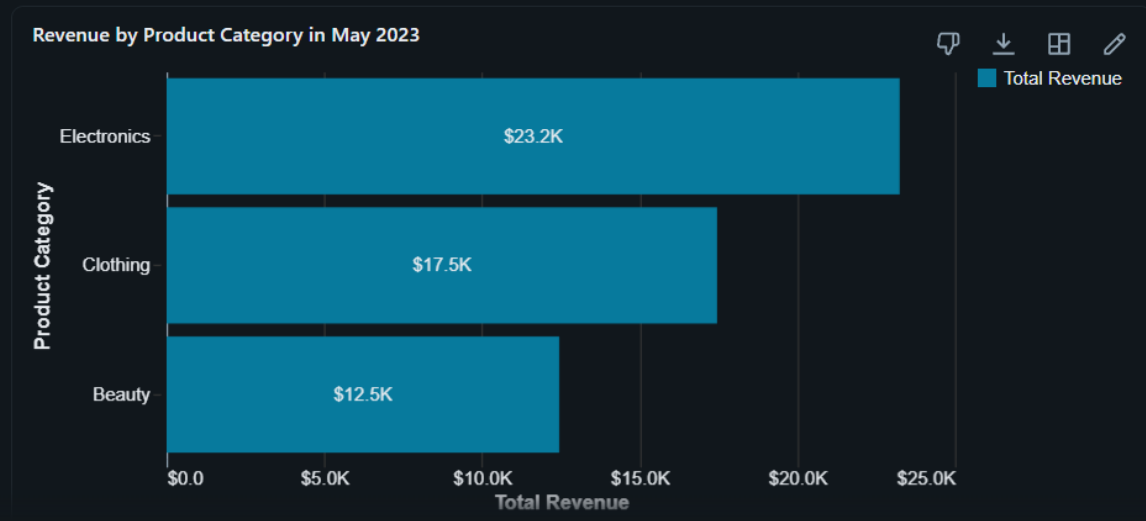
Electronics Dominated May with \$23,245 in Revenue

In May 2023 (the highest quantity month), Electronics generated the most revenue at \$23,245, representing 43.7% of the month's total revenue. ²

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Show code



May 2023 Category Breakdown

1. **Electronics:** \$23,245 revenue (97 units, 40 transactions, \$581.13 avg/transaction)
2. **Clothing:** \$17,455 revenue (97 units, 37 transactions, \$471.76 avg/transaction)
3. **Beauty:** \$12,450 revenue (65 units, 28 transactions, \$444.64 avg/transaction) ²

Key Insights

May's success driven by Electronics: Electronics not only led revenue in May but also had the **highest average transaction value** at \$581.13, indicating customers were purchasing high-value electronic items during this peak month.

Equal volume, different value: Electronics and Clothing both sold 97 units in May, but Electronics generated **33% more revenue** (\$23,245 vs \$17,455), highlighting the higher price points of electronic products.

Consistent pattern: May was the top performer in both **quantity sold** (259 units) and **total revenue** (\$53,150), while September was the worst in both metrics (170 units, \$23,620 revenue), showing that volume and revenue move together.

Business Recommendations

1. **Replicate May's Success:** Analyze what made May successful:
 - Were there specific promotions or marketing campaigns?
 - Did product launches drive Electronics sales?
 - Consider running similar campaigns in low-performing months
2. **Electronics Focus:** Since Electronics drove May's peak performance with high-value transactions:
 - Stock premium electronics during traditionally strong months
 - Create Electronics-focused promotions during weak months to boost revenue
 - Bundle high-margin electronics with other categories
3. **Address September's Weakness:** With both lowest quantity and revenue:
 - Implement the seasonal promotion strategies suggested earlier
 - Focus on Electronics promotions (your revenue driver) during September
 - Consider "End of Summer" or "Back to School Tech" campaigns

9. Looking at both revenue and units sold, which product category performed strongly? Explain your answer.

Agent answer:

Electronics Performed Strongest Overall

Looking at both revenue and units sold, **Electronics** is the strongest performing product category. Here's why:

Performance Summary

Electronics:

- Revenue: \$156,905 (34.41% market share) - **#1 Highest**
- Units Sold: 849 units (33.77% market share) - **#2**
- Average Price per Unit: \$184.81 - **Highest**
- Transactions: 342 [1](#)

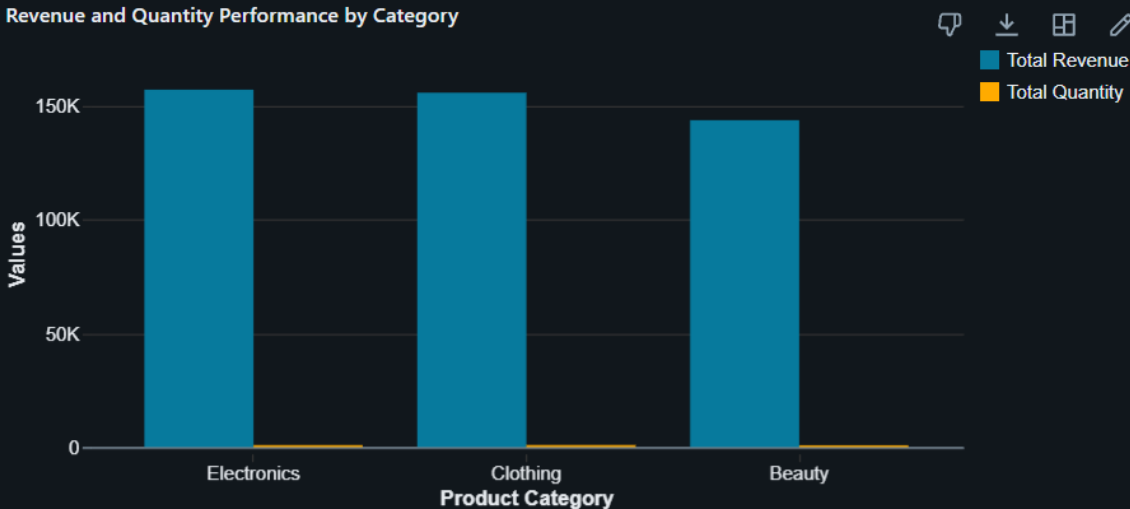
Clothing:

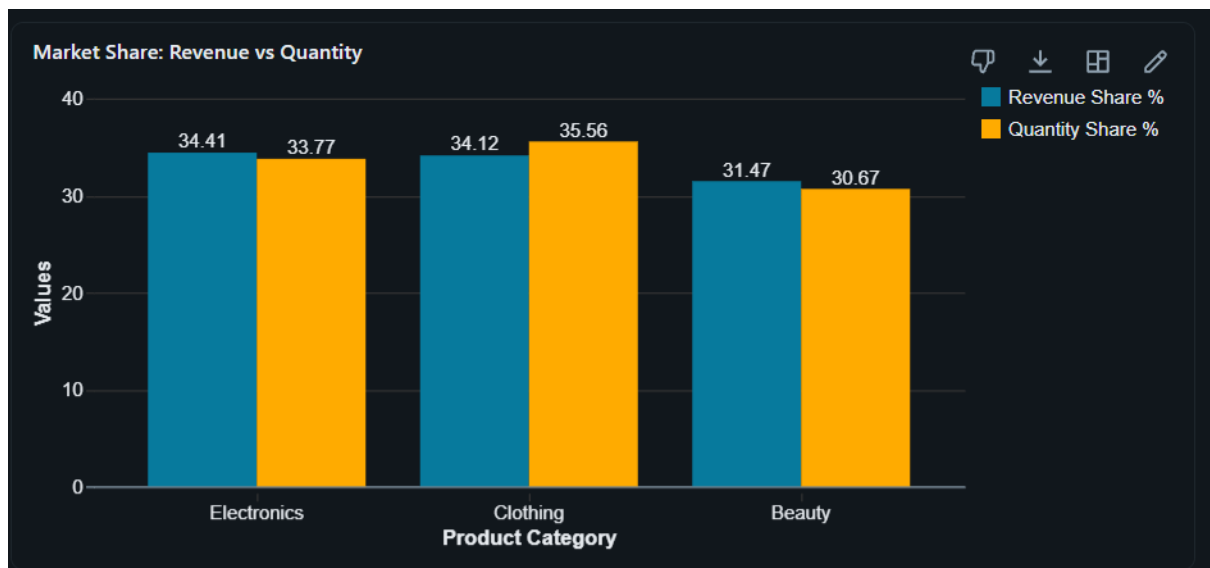
- Revenue: \$155,580 (34.12% market share) - **#2**
- Units Sold: 894 units (35.56% market share) - **#1 Highest**
- Average Price per Unit: \$174.03 - **Lowest**
- Transactions: 351 [1](#)

Beauty:

- Revenue: \$143,515 (31.47% market share) - **#3**
- Units Sold: 771 units (30.67% market share) - **#3**
- Average Price per Unit: \$186.14 - **Highest**
- Transactions: 307 [1](#)

Revenue and Quantity Performance by Category





Why Electronics Performed Strongest

1. Revenue Leadership with Efficiency

Electronics generated the **highest total revenue** (\$156,905) while selling **fewer units** than Clothing (849 vs 894 units). This demonstrates superior revenue efficiency—Electronics generates more money with less volume. ¹

2. Balanced Market Position

Electronics holds the **most balanced position** across both metrics:

- **34.41% revenue share** (highest)
- **33.77% quantity share** (middle)

The gap between revenue and quantity share is only 0.64 percentage points, showing Electronics captures proportional value for its volume. ²

In contrast:

- **Clothing** has a 1.44 percentage point gap (35.56% quantity vs 34.12% revenue)—selling more units but capturing less revenue per unit
- **Beauty** has a 0.80 percentage point gap (30.67% quantity vs 31.47% revenue)—slightly better revenue capture but overall lower performance

3. Premium Pricing Power

Electronics commands a **high average price per unit** of \$184.81, nearly matching Beauty's \$186.14 (the highest) and significantly above Clothing's \$174.03. This pricing power, combined with strong volume, drives superior revenue performance. ¹

4. Strong Transaction Value

Electronics achieves an **average revenue per transaction** of \$458.79, positioning it between Clothing (\$443.25) and Beauty (\$467.48). With 342 transactions, Electronics maintains consistent high-value sales. ¹

5. Consistent Performance Across Time

Throughout our previous analyses, Electronics consistently showed strength:

- **Dominated May** (the highest revenue month) with \$23,245—43.7% of that month's revenue
- Maintained strong performance even in lower months
- Showed resilience across different customer segments

Why Not Clothing?

While Clothing sold the **most units** (894), it performed **second** overall because:

1. **Lower revenue per unit:** At \$174.03 per unit, Clothing generates the least revenue per item sold
2. **Volume without premium value:** Selling 45 more units than Electronics but generating \$1,325 less revenue shows a value capture problem
3. **Market share imbalance:** 35.56% of quantity but only 34.12% of revenue—giving away 1.44 percentage points of potential revenue ²

Why Not Beauty?

Beauty ranks **third** in both metrics:

1. **Lowest volume:** Only 771 units sold (30.67% market share)
2. **Lowest revenue:** \$143,515 (31.47% market share)
3. **Despite highest price per unit** (\$186.14), Beauty couldn't overcome its volume disadvantage ¹

Conclusion

Electronics is the strongest performer because it achieves the optimal balance: **#1 in revenue generation** while maintaining competitive volume (#2 in units sold) and premium pricing. It captures the highest market share in the metric that matters most—revenue—while efficiently converting fewer transactions into maximum value.

Electronics demonstrates the business principle that **revenue efficiency matters more than raw volume**. Selling slightly fewer units at higher prices with strong customer demand creates a more profitable and sustainable business model than high-volume, lower-margin sales.

10. What is the total quantity sold?

Agent answer:

Total Quantity Sold: 2,514 Units

The total quantity sold across all transactions is **2,514 units**. ¹

This breaks down by product category as:

- **Clothing:** 894 units (35.6%)
- **Electronics:** 849 units (33.8%)
- **Beauty:** 771 units (30.7%)